Unit - V

THEORY OF GAMES

Introduction:

Game theory was developed for the purpose of analyzing competitive situations involving conflicting interests. In other words, game theory is used for decision making under conflicting situations where there are one or more opponents (i.e., players). For example, chess, poker, etc., are the games which have the characteristics of a competition and are played according to definite rules. Game theory provides solutions to such games, assuming that each of the players wants to maximize his profits and minimize his losses.

Categories:

The game theory models can be classified into several categories. Some important categories are listed below.

i. **Two-person & N-person games:** If the number of players is two, it is known as two-person game. On the other hand, if the number of players is N, it is known as N-person game.

ii. **Zero sum & Non-zero sum game:** In a zero sum game, the sum of the points won equals the sum of the points lost, i.e., one player wins at the expense of the other. To the contrary, if the sum of gains or losses is not equal to zero, it is either positive or negative,then it is known as non-zero sum game. An example of non-zero sum game is the case of two competing firms each with a choice regarding its advertising campaign. In such a situation, both the firms may gain or loose, though their gain or loss may not be equal.

iii. **Games of Perfect and Imperfect information:** If the strategy of a player can be discovered by his competitor, then it is known as a perfect information game. In case of imperfect information games no player has complete information and tries to guess the real situation.

iv. **Pure & Mixed strategy games:** If the players select the same strategy each time, then it is referred to as pure strategy games. If a player decides to choose a course of action for each play in accordance with some particularly probability distribution, it is called mixedstrategy game.

Assumptions of game theory:

i. There is finite number of competitors (players).
ii. The players act reasonably.
iii. Every player strives to maximize gains and minimize losses.
iv. Each player has finite number of possible courses of action.
v. The choices are assumed to be made simultaneously, so that no player knows his opponent's choice until he has decided his own course of action.
vi. The pay-off is fixed and predetermined.
vii. The pay-offs must represent utilities.

Basic Terminology:

i. **Player:** Each participant (interested party) is called a player.

ii. **Strategy:** The strategy of a player is the predetermined rule by which a player decides his course of action from the list of courses of action during the game. A strategy may be of two types:

   a. **Pure strategy:** It is a decision, in advance of all plays, always to choose a particular course of action. In other words, if the best strategy for each player is

to play one particular strategy throughout the game, it is called pure strategy.

b. **Mixed strategy:** It is a decision, in advance of all plays, to choose a course of action for each play in accordance with some particular probability distribution. In other words, if the optimal plan for each player is to employ different strategies at different times, we call it mixed strategy.

iii. **Optimal strategy:** The course of action which maximizes the profit of a player or minimizes his loss is called an optimal strategy.

iv. **Saddle point:** A saddle point is an element of the matrix that is both the smallest element in its row and the largest element in its column. Furthermore, saddle point is also regarded as an equilibrium point in the theory of games.

v. **Pay-off:** The outcome of playing the game is called pay-off.

vi. **Pay-off Matrix:** It is a table showing the outcomes or payoffs of different strategies of the game.

vii. **Value of the Game:** It refers to the expected outcome per play, when players follow their optimal strategy. It is generally denoted by V.

Advantages of Game Theory:

i. Game theory gives insight into several less-known aspects, which arise in situations of conflicting interests. For example, it describes and explains the phenomena of bargaining and coalition-formation.

ii. Game theory develops a framework for analyzing decision making in such situations where interdependence of firms is considered.

iii. At least in two-person zero-sum games, game theory outlines a scientific quantitative technique that can be used by players to arrive at an optimal strategy.

Limitations of Game Theory:

i. The assumption that players have the knowledge about their own pay-offs and pay-offs of others is not practical.

ii. The techniques of solving games involving mixed strategies particularly in case of large pay-off matrix is very complicated.

iii. All the competitive problems cannot be analyzed with the help of game theory.

Pure Strategy: The simplest type of game is one where the best strategies for both players are pure strategies. This is the case if and only if, the pay-off matrix contains a saddle point.

Problem: For the game with payoff matrix find the optimal solution.

		Player B				
		I	II	III	IV	V
Player A	I	-2	0	0	5	3
	II	4	2	1	3	2
	III	4	-3	0	-2	6
	IV	5	3	-4	2	-6

Solution:

We use the maximin (minimax) principle to analyze the game.

		Player B					Minimum
		I	II	III	IV	V	
Player A	I	-2	0	0	5	3	-2
	II	4	2	1	3	2	1
	III	4	-3	0	-2	6	-3
	IV	5	3	-4	2	-6	-6
Maximum		5	3	1	5	6	

Select minimum from the maximum of columns. Minimax = 1
Player A will choose II strategy, which yields the maximum payoff of 1. Select maximum from the minimum of rows. Maximin = 1
Similarly, player B will choose III strategy.
Since the value of maximin coincides with the value of the minimax, therefore, saddle point (equilibrium point) = 1.
The optimal strategies for both players are: Player A must select II strategy and player B mustselect III strategy. The value of game is 1, which indicates that player A will gain 1 unit and player B will sacrifice 1 unit.

Mixed Strategy: In situations where a saddle point does not exist, the maximin(minimax) principle for solving a game problem breaks down.

Problem: Two companies A and B are competing for the same product. Their different strategiesare given in the following pay-off matrix:.

		Company B		
		I	II	III
Company A	I	-2	14	-2
	II	-5	-6	-4
	III	-6	20	-8

Determine the optimal strategies for both the companies.

Solution:
First, apply the maximin (minimax) principle to analyze the game.

		Company B			Minimum
		I	II	III	
Company A	I	-2	14	-2	-2
	II	-5	-6	-4	-6
	III	-6	20	-8	-8
Maximum		-2	20	-2	

Minimax = -2 & Maximin = -2
There are two elements whose value is –2. Hence, the solution to such a game is not unique.

Algebraic Method: Consider the zero sum two-person game given below:

		Player B	
		I	II
Player A	I	a	b
	II	c	d

The solution of the game is:

A play's (p, 1 - p)
Where p = (d - c) / [(a + d) - (b + c)]B
play's (q, 1 - q)
Where q = (d - b) / [(a + d) - (b + c)]
Value of the game, V = (ad - bc) / [(a + d) - (b + c)]

Problem: Consider the game of matching coins. Two players, A & B, put down a coin. If coins match (i.e., both are heads or both are tails) A gets rewarded, otherwise B. However, matching on heads gives a double premium. Obtain the best strategies for both players and the value of the game.

		Player B	
		I	II
Player A	I	2	-1
	II	-1	1

Solution: This game has no saddle point.A play's (p, 1 - p):

where p = (d - c) / [(a + d) - (b + c)]
= [1 - (-1)] / [(2 + 1) - (-1 - 1)]
= 2/5

1 – p = 3/5

B play's (q, 1 - q):

Where q = (d - b) / [(a + d) - (b + c)]
= [1 - (-1)] / [(2 + 1) - (-1 - 1)]
= 2/5

1 –q = 3/5

Value of the game

V = (ad - bc) / [(a + d) - (b + c)]
= 2 x 1 - (-1) x (-1) / (2 + 1) - (-1 - 1)
= 1/5

Problem: Solve the game whose payoff matrix is given below:

		Player B	
		I	II
Player A	I	1	7
	II	6	2

Solution: This game has no saddle point.

p= (d - c) / [(a + d) - (b + c)]
= [2 - 6] / [(1 + 2) - (7 + 6)]
= 2/51
p = 3/5

q = (d - b) / [(a + d) - (b + c)]
= [2 - 7] / [(1 + 2) - (7 + 6)]
= 2/51
q = 1/2

Value of the game

V = (ad - bc) / [(a + d) - (b + c)]
= 1 x 2 - 7 x 6 / (1 + 2) - (7 + 6)
= 4

Dominance Property:

The principle of dominance states that if one strategy of a player dominates over the other strategy in all conditions then the later strategy can be ignored. A strategy dominates over the other only if it is preferable over other in all conditions. The concept of dominance is especially useful for the evaluation of two-person zero-sum games where a saddle point does not exist.

Rules:

i. If all the elements of a column (say i th column) are greater than or equal to the corresponding elements of any other column (say j th column), then the i th column is dominated by the j th column and can be deleted from the matrix.

ii. If all the elements of a row (say i th row) are less than or equal to the corresponding elements of any other row (say j th row), then the i th row is dominated by the j th row and can be deleted from the matrix.

iii. In the pay off matrix of player A, if each of the average sum of the entries of any two rows is grater than or equal to the corresponding entry of third row, then third row is dominated by previous two rows, then third row can be deleted.

iv. In the pay off matrix of player B, if average sum of the entries of any two columns is lesser than equal to the corresponding entry of third column, then third column is dominated by sum of two previous columns, then third column is deleted.

Problem: Use the concept of dominance to solve the following game:

		Player B			
		I	II	III	IV
Player A	I	3	5	4	2
	II	5	6	2	4
	III	2	1	4	0
	IV	3	3	5	2

Solution:

		Player B				Minimum
		I	II	III	IV	
Player A	I	3	5	4	2	2
	II	5	6	2	4	2
	III	2	1	4	0	0
	IV	3	3	5	2	2
Maximum		5	6	5	4	

There is no saddle point in this game.

If a column is greater than another column (compare corresponding elements), then delete that column. Here, I and II column are greater than the IV column. So, player B has no incentive in using his I and II course of action.

		Player B	
		III	IV
Player A	I	4	2
	II	2	4
	III	4	0
	IV	5	2

If a row is smaller than another row (compare corresponding elements), then delete that row. Here, I and III row are smaller than IV row. So, player A has no incentive in using his I and III course of action.

		Player B	
		III	IV
Player A	II	2	4
	IV	5	2

Value of the game is found by algebraic methodp = (d - c) / [(a + d) - (b + c)]

= [2 - 5] / [(2 + 2) - (4 + 5)]

= 3/51 – p = 2/5

q = (d - b) / [(a + d) - (b + c)]
= [2 - 4] / [(2 + 2) - (4 + 5)]
= 2/51 –q = 3/5
Value of the game
V = (ad - bc) / [(a + d) - (b + c)]
= 2 x 2 - 4 x 5 / (2 + 2) - (4 + 5)
= - 16 / - 16
= 1

Graphical Method:
The method discussed in the previous section is feasible when the value of n is small, because the larger value of n will yield a larger number of 2 x 2 sub-games. In this section, we discuss another method for solving 2 X n games. This method can only be used in games with no saddle point, and having a pay-off matrix of type n x 2 or 2 x n.

Problem: Use graphical method in solving the following game and find the value of the game.

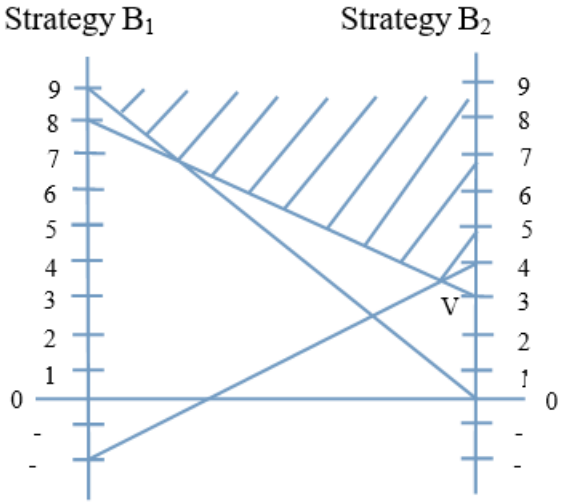
		Player B	
		B_1	B_2
Player A	A_1	-2	4
	A_2	8	3
	A_3	9	0

Solution: The game does not have a saddle point as shown in the following table.

		Player B		Minimum	Probability
		B_1	B_2		
Player A	A_1	-2	4	-2	q_1
	A_2	8	3	3	q_2
	A_3	9	0	0	q_3
Maximum		9	4		
Probability		p_1	p_2		

Maximin = 4 & Minimax = 3

First, draw two parallel lines 1 unit distance apart and mark a scale on each. The two parallel lines represent strategies of player B. If player A selects strategy A_1, player B can win -2 (i.e., loose 2 units) or 4 units depending on B' s selection of strategies. The value -2 is plotted along the vertical axis under strategy B_1 and the value 4 is plotted along the vertical axis under strategy B_2 . A straight line joining the two points is then drawn. Similarly, we can plot strategies A_2 and A_3 also.



The lowest point V in the shaded region indicates the value of game. From the above figure,

the value of the game is 3.4 units. Likewise, we can draw a graph for player B.

The point of optimal solution (i.e., maximin point) occurs at the intersection of two lines:
$E1 = -2p_1 + 4p_2$ and
$E2 = 8p_1 + 3p_2$
Comparing the above two equations, we have
$-2p_1 + 4p_2 = 8p_1 + 3p_2$
Substituting p 2 = 1 - p 1
$-2p_1 + 4(1 - p_1) = 8p_1 + 3(1 - p_1)p_1 = 1/11$
$p_2 = 10/11$
Substituting the values of p_1 and p_2 in equation
$E_1V = -2\ (1/11) + 4\ (10/11) = 3.4$ units